

Model, estimation and control of the air-ground mobile cable-driven parallel robot

Derivations for the C++ core (`core/`)

This document derives the model that `core/` implements, from the single unit (one cable, one winch, one quadrotor, one ground robot) to the coupled machine, and then the estimators and controllers built on it. Each section names the function that implements it. Numbers quoted in Section 9 come from `tests/core_test.py` and `tests/isaac_physics_test.py`.

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1 Frames and notation

World frame $\mathcal{W}$: z up, ground at $z = 0$, $\mathbf{e}_3 = (0, 0, 1)^\top$. Every rigid body carries a frame at its centre of mass; $\mathbf{R} \in \text{SO}(3)$ maps body to world. $\boldsymbol{\Omega}$ is the angular velocity in the body frame, so $\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\Omega}]_\times$, with $[\mathbf{a}]_\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$. Indices: cables $i = 1..8$; cables 1..4 go from the platform's top corners to the quadrotors, 5..8 from the bottom corners to the ground robots.

$\mathbf{p}, \mathbf{v}, \mathbf{R}, \boldsymbol{\Omega}$	platform position, velocity, attitude, body rate
$m, \mathbf{J}$	platform mass and inertia (3 kg, cube of side 0.30 m)
$\mathbf{b}_i$	attachment point of cable i in the platform frame (cube corners)
$\mathbf{a}_i, \dot{\mathbf{a}}_i$	anchor (fairlead) position and velocity of cable i
$d_i, \mathbf{u}_i$	chord length and unit vector from attachment point to anchor
L_i, T_i	unstretched cable length, cable tension
EA, c_{EA}, ρ	cable axial stiffness, damping, linear density

2 Unit models

2.1 Cable

Kinematics (`cable_geometry`, `cable_rates`). With $\mathbf{r}_i = \mathbf{R}\mathbf{b}_i$,

$$\mathbf{d}_i = \mathbf{a}_i - \mathbf{p} - \mathbf{r}_i, \quad d_i = \|\mathbf{d}_i\|, \quad \mathbf{u}_i = \mathbf{d}_i/d_i, \quad \dot{\mathbf{d}}_i = \mathbf{u}_i^\top (\dot{\mathbf{a}}_i - \mathbf{v} - \boldsymbol{\omega} \times \mathbf{r}_i), \quad \boldsymbol{\omega} = \mathbf{R}\boldsymbol{\Omega}. \quad (1)$$

Elasticity (`cable_tension`). The cable is a tension-only Kelvin–Voigt element whose stiffness and damping scale with $1/L_i$:

$$T_i = \begin{cases} \max\left(0, \frac{EA}{L_i}(d_i - L_i) + \frac{c_{EA}}{L_i}(\dot{d}_i - \dot{L}_i)\right) & d_i > L_i \\ 0 & d_i \leq L_i \end{cases} \quad (2)$$

Statically $d_i = L_i(1 + T_i/EA)$, which is how a winch encoder (L_i) and a load cell (T_i) together measure the chord (`chord_length`).

Sag. A cable of weight per length ρg at elevation β_i ($\sin \beta_i = u_{i,z}$) carries a normal load $w_n = \rho g \cos \beta_i$. For small sag the profile is a parabola and the chord is shorter than the stretched length $s_i = L_i(1 + T_i/EA)$ by

$$d_i = s_i \left(1 - \frac{(w_n s_i)^2}{24 T_i^2}\right). \quad (3)$$

For the 2 mm rope ($\rho = 2.6$ g/m) at 5 m and 15 N this is 0.04 mm, so the simulation uses straight cables with half of each cable's weight lumped at its ends; the correction is available in the observer (`observer.sag`).

2.2 Winch

Drum of radius r_d , reflected inertia J_w , viscous and Coulomb friction, in cable coordinates ($m_w = J_w/r_d^2 = 5$ kg) (`winch_step`):

$$m_w \ddot{L}_i = F_i + T_i - c_v \dot{L}_i - F_c \operatorname{sgn}(\dot{L}_i), \quad |F_i| \leq F_{\max} \left(1 - \frac{|\dot{L}_i|}{v_{\max}}\right) \text{ when driving.} \quad (4)$$

The motor force is one of two servos, both with the tension feed-forward T_i^{ff} :

$$\text{length mode: } F_i = -T_i^{\text{ff}} + m_w \left[\omega_w^2 (L_i^c - L_i) + 2\zeta_w \omega_w (\dot{L}_i^c - \dot{L}_i) \right], \quad (5)$$

$$\text{tension mode: } F_i = -T_i^{\text{ff}} + c_w (\dot{L}_i^c - \dot{L}_i), \quad c_w = 2\zeta_w \omega_w m_w = 450 \text{ N s/m.} \quad (6)$$

In tension mode at steady speed $T_i = T_i^{\text{ff}} + c_w (\dot{L}_i - \dot{L}_i^c) + c_v \dot{L}_i + F_c \operatorname{sgn} \dot{L}_i$: the winch is a force source in series with a damper referenced to the commanded rate.

2.3 Quadrotor

Mass m_q , inertia $\mathbf{J}_q$, four rotors with thrust $f_j = k_f \varpi_j^2$, first-order speed dynamics $\dot{\varpi}_j = (\varpi_j^c - \varpi_j)/\tau_m$, and mixer (`quad_mixer`)

$$\begin{pmatrix} f \\ \mathbf{M} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ y_1 & y_2 & y_3 & y_4 \\ -x_1 & -x_2 & -x_3 & -x_4 \\ -\kappa s_1 & -\kappa s_2 & -\kappa s_3 & -\kappa s_4 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix}. \quad (7)$$

The cable pulls the fairlead (offset $\mathbf{r}_a$ from the centre of mass) with $-T\mathbf{u}$:

$$m_q \dot{\mathbf{v}}_q = f \mathbf{R}_q \mathbf{e}_3 - m_q g \mathbf{e}_3 - T\mathbf{u} - \frac{1}{2} \rho g L \mathbf{e}_3 + \mathbf{F}_{\text{drag}}, \quad (8)$$

$$\mathbf{J}_q \dot{\boldsymbol{\Omega}}_q = \mathbf{M} - \boldsymbol{\Omega}_q \times \mathbf{J}_q \boldsymbol{\Omega}_q + \mathbf{r}_a \times \mathbf{R}_q^\top (-T\mathbf{u}), \quad (9)$$

with $\mathbf{F}_{\text{drag}} = -\mathbf{R}_q \operatorname{diag}(k_d, k_d, 0) \mathbf{R}_q^\top \mathbf{v}_q - c_d \|\mathbf{v}_q\| \mathbf{v}_q$ (`quad_wrench`).

2.4 Ground robot

Differential drive, pose (x, y, θ) , forward speed v , yaw rate ω , wheel radius r , track b . Rolling without slipping gives $\dot{x} = v \cos \theta$, $\dot{y} = v \sin \theta$, $\dot{\theta} = \omega$ and wheel speeds $\varpi_{l,r} = (v \mp \frac{b}{2} \omega)/r$. The wheels are velocity servos, $\tau_{l,r} = k_w (\varpi_{l,r}^c - \varpi_{l,r})$, $|\tau| \leq \tau_{\max}$. The fairlead sits on the axle axis at height h , so the cable force $\mathbf{F} = -T\mathbf{u}$ has no yaw moment:

$$\left(m_g + \frac{2J_{wh}}{r^2}\right) \dot{v} = \frac{\tau_l + \tau_r}{r} + \mathbf{F} \cdot \mathbf{h}, \quad \left(I_z + \frac{J_{wh} b^2}{2r^2}\right) \dot{\omega} = \frac{(\tau_r - \tau_l) b}{2r}, \quad (10)$$

with $\mathbf{h} = (\cos \theta, \sin \theta, 0)^\top$. The lateral component $\mathbf{F} \cdot \mathbf{h}_\perp$ and the drive force are carried by tyre friction; the model reports the friction demand over μN , $N = m_g g + F_z$ (`Simulator::traction_use`).

2.5 Platform

Newton–Euler about the centre of mass with the external wrench $(\mathbf{f}_e, \boldsymbol{\tau}_e)$ of the tool and payload:

$$m \dot{\mathbf{v}} = \sum_i T_i \mathbf{u}_i - m g \mathbf{e}_3 + \mathbf{f}_e, \quad \mathbf{J} \dot{\boldsymbol{\Omega}} = \mathbf{R}^\top \left(\sum_i \mathbf{r}_i \times T_i \mathbf{u}_i + \boldsymbol{\tau}_e \right) - \boldsymbol{\Omega} \times \mathbf{J} \boldsymbol{\Omega}. \quad (11)$$

3 The coupled machine

3.1 Structure matrix

Collecting (11), the cables apply the wrench $\mathbf{W}\mathbf{t}$ with

$$\mathbf{W}(\mathbf{p}, \mathbf{R}, \mathbf{a}) = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_8 \\ \mathbf{r}_1 \times \mathbf{u}_1 & \cdots & \mathbf{r}_8 \times \mathbf{u}_8 \end{pmatrix} \in \mathbb{R}^{6 \times 8}, \quad \mathbf{t} \geq 0. \quad (12)$$

By (1), $\dot{\mathbf{d}} = -\mathbf{W}^\top \boldsymbol{\xi} + \mathbf{U}^\top \dot{\mathbf{a}}$ with the platform twist $\boldsymbol{\xi} = (\mathbf{v}, \boldsymbol{\omega})$: $-\mathbf{W}^\top$ is the inverse-kinematics Jacobian. Eight cables on six degrees of freedom leave a two-dimensional null space for internal tension.

3.2 State and dimension

The plant state (`Simulator`) is

$$\underbrace{(\mathbf{p}, \mathbf{v}, \mathbf{R}, \boldsymbol{\Omega})}_{12} \oplus \underbrace{4 \times (\mathbf{p}_q, \mathbf{v}_q, \mathbf{R}_q, \boldsymbol{\Omega}_q, \boldsymbol{\varpi})}_{4 \times 16} \oplus \underbrace{4 \times (x, y, \theta, v, \omega)}_{4 \times 5} \oplus \underbrace{(\mathbf{L}, \dot{\mathbf{L}})}_{16}$$

i.e. 112 states, coupled only through the eight cable tensions (2): each tension depends on the platform pose, one robot's pose and one winch. The inputs are 16 rotor speed commands, 8 wheel speed commands and 8 winch commands ($L^c, \dot{L}^c, T^{\text{ff}}$, mode). Integration is semi-implicit Euler at 1 kHz, the same step as PhysX.

3.3 Tension bounds from the robots' limits

Each robot holds exactly one cable, so its limits are bounds on that one tension.

Quadrotor (`drone_tension_cap`). In equilibrium (8) needs thrust $\mathbf{f} = m_q g \mathbf{e}_3 + T\mathbf{u}$. Tilt $\leq \theta_{\max}$ and $\|\mathbf{f}\| \leq f_{\max}$ give

$$T \leq \frac{m_q g \tan \theta_{\max}}{u_{xy} - u_z \tan \theta_{\max}} \quad (\text{if } u_{xy} > u_z \tan \theta_{\max}), \quad T \leq -m_q g u_z + \sqrt{(m_q g u_z)^2 - (m_q g)^2 + f_{\max}^2}, \quad (13)$$

with $u_{xy} = \|(u_x, u_y)\|$.

Ground robot (`ugv_tension_cap`). The cable pulls with Tu_{xy} horizontally and $-Tu_z$ vertically. $\mathbf{u}$ points from the platform to the anchor, so for a ground robot $u_z < 0$ and the cable lifts it: the normal force is $N = m_g g + Tu_z$. Sliding: $Tu_{xy} \leq \mu(m_g g + Tu_z)$. Tipping about the edge of the support polygon at distance a (inradius of the wheel-caster rhombus, 0.148 m) with the fairlead at height h : $Tu_{xy}h \leq (m_g g + Tu_z)a$. Hence

$$T \leq \frac{\mu m_g g}{u_{xy} - \mu u_z}, \quad T \leq \frac{m_g g a}{u_{xy} h - u_z a}. \quad (14)$$

For the default layout the tipping bound is 63 N at 80% margin, just above the 60 N winch limit; the sliding bound is about 130 N.

3.4 Tension distribution

Given the wrench $\mathbf{w}$ the cables must supply, the tensions solve (`tension_distribution`)

$$\min_{\mathbf{t}} \lambda \|\mathbf{t} - t_{\text{ref}} \mathbf{1}\|^2 + \lambda \kappa \|\mathbf{t} - \mathbf{t}^-\|^2 + \|\mathbf{W}\mathbf{t} - \mathbf{w}\|^2 \quad \text{s.t.} \quad \mathbf{t}_{\min} \leq \mathbf{t} \leq \mathbf{t}_{\max}(\mathbf{u}), \quad (15)$$

where $\mathbf{t}_{\max}$ is the smaller of the winch limit and (13), (14). It is a strictly convex box-constrained QP in eight variables, solved exactly by a primal active-set method (`solve_box_qp`): minimise over the free set, step to the first bound hit, release the bound whose multiplier has the wrong sign, stop when the projected gradient vanishes. Without active bounds and $\kappa = 0$ the solution is the damped pseudo-inverse $\mathbf{t} = \mathbf{t}_{\text{ref}} + \mathbf{W}^\top (\mathbf{W}\mathbf{W}^\top + \lambda \mathbf{I})^{-1} (\mathbf{w} - \mathbf{W}\mathbf{t}_{\text{ref}})$. The damping follows the conditioning of $\mathbf{W}$ (`adaptive_lambda`):

$$\lambda = \lambda_{\min} + (1 - (\sigma_{\min}/\sigma_\epsilon)^2)(\lambda_{\max} - \lambda_{\min}) \text{ for } \sigma_{\min}(\mathbf{W}) < \sigma_\epsilon, \quad \lambda = \lambda_{\min} \text{ otherwise,} \quad (16)$$

so the wrench is met to 10^{-6} in normal layouts ($\sigma_{\min} = 0.27$ for the crossed layout) and is not forced through vanishing moment arms near a singular one.

4 Sensors and estimation

Sensor	Measures	Rate	Noise (1σ)
Drone IMU	specific force, body rate	500 Hz	$6.9 \times 10^{-4} \text{ m/s}^2/\sqrt{\text{Hz}}$, $4.9 \times 10^{-5} \text{ rad/s}/\sqrt{\text{Hz}}$, biases
Drone fix	position, heading	50 Hz	5 mm, 0.5°
Wheel encoders	wheel rates	100 Hz	0.02 rad/s
Ground-robot fix (lidar)	x, y, θ	15 Hz	5 mm, 0.3°
Winch encoder	$L_i, \dot{L}_i$	1 kHz	0.03 mm, 1 mm/s
Winch load cell	T_i	1 kHz	0.1 N
Robot camera + AprilTag	platform pose in the camera frame	30 Hz	3 mm, 0.3°

The measurements are generated from the true simulator state by **SensorSuite** (rates, white noise, IMU bias and bias random walk). The camera images and lidar scans themselves are not processed in this loop.

4.1 Quadrotor: error-state Kalman filter

Nominal state $(\mathbf{p}_q, \mathbf{v}_q, \mathbf{R}_q, \mathbf{b}_g, \mathbf{b}_a)$, propagated with the IMU sample $(\mathbf{a}_m, \boldsymbol{\omega}_m)$: $\dot{\mathbf{p}}_q = \mathbf{v}_q$, $\dot{\mathbf{v}}_q = \mathbf{R}_q(\mathbf{a}_m - \mathbf{b}_a) - g\mathbf{e}_3$, $\dot{\mathbf{R}}_q = \mathbf{R}_q[\boldsymbol{\omega}_m - \mathbf{b}_g]_\times$. The error $\delta\mathbf{x} = (\delta\mathbf{p}, \delta\mathbf{v}, \delta\boldsymbol{\theta}, \delta\mathbf{b}_g, \delta\mathbf{b}_a)$, with $\mathbf{R}_q \leftarrow \mathbf{R}_q \exp[\delta\boldsymbol{\theta}]_\times$, obeys

$$\delta\dot{\mathbf{p}} = \delta\mathbf{v}, \quad \delta\dot{\mathbf{v}} = -\mathbf{R}_q[\mathbf{a}_m - \mathbf{b}_a]_\times \delta\boldsymbol{\theta} - \mathbf{R}_q \delta\mathbf{b}_a, \quad \delta\dot{\boldsymbol{\theta}} = -[\boldsymbol{\omega}_m - \mathbf{b}_g]_\times \delta\boldsymbol{\theta} - \delta\mathbf{b}_g. \quad (17)$$

The position fix updates with $\mathbf{H} = [\mathbf{I} \ \mathbf{0} \ \mathbf{0} \ \mathbf{0} \ \mathbf{0}]$; the heading $\psi = \text{atan2}(R_{21}, R_{11})$ with $\partial\psi/\partial\delta\boldsymbol{\theta} = (R_{11}\mathbf{A}_2 - R_{21}\mathbf{A}_1)/(R_{11}^2 + R_{21}^2)$, $\mathbf{A} = -\mathbf{R}_q[\mathbf{e}_1]_\times$ (**DroneEskf**). The fairlead is $\mathbf{a} = \mathbf{p}_q + \mathbf{R}_q \mathbf{r}_a$.

4.2 Ground robot: odometry + pose fix

(x, y, θ) is propagated with the encoder odometry $v = \frac{r}{2}(\varpi_l + \varpi_r)$, $\omega = \frac{r}{b}(\varpi_r - \varpi_l)$ and corrected by the pose fix (**UgvEKF**); the anchor is (x, y, h) .

4.3 Platform: tension-driven observer

The platform carries no inertial sensor. Its observer (**PlatformObserver**) uses the model (11) itself as the predictor, driven by the *measured* tensions, and adds the unknown external wrench as a random-walk state: nominal state $(\mathbf{p}, \mathbf{v}, \mathbf{R}, \boldsymbol{\Omega}, \mathbf{d}_f, \mathbf{d}_\tau)$,

$$\dot{\mathbf{v}} = \frac{1}{m} \left(\sum_i T_i \mathbf{u}_i + \mathbf{f}_k + \mathbf{d}_f \right) - g\mathbf{e}_3, \quad \mathbf{J}\dot{\boldsymbol{\Omega}} = \sum_i \mathbf{b}_i \times \mathbf{R}^\top T_i \mathbf{u}_i + \mathbf{R}^\top \boldsymbol{\tau}_k + \mathbf{d}_\tau - \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega}, \quad (18)$$

with $(\mathbf{f}_k, \boldsymbol{\tau}_k)$ the known tool wrench (wrist load cell). Because T_i is measured, the stiff elastic law (2) does not appear in the predictor; only the direction $\mathbf{u}_i$ depends on the state: $\partial(T_i \mathbf{u}_i)/\partial \mathbf{x}_{a,i} =$

$-\frac{T_i}{d_i} \mathbf{P}_i$, $\mathbf{P}_i = \mathbf{I} - \mathbf{u}_i \mathbf{u}_i^\top$, the geometric stiffness. With $\delta \mathbf{x}_{a,i} = \delta \mathbf{p} - \mathbf{R}[\mathbf{b}_i]_\times \delta \boldsymbol{\theta}$ the error dynamics are

$$m \delta \dot{\mathbf{v}} = - \sum_i \frac{T_i}{d_i} \mathbf{P}_i \delta \mathbf{p} + \sum_i \frac{T_i}{d_i} \mathbf{P}_i \mathbf{R}[\mathbf{b}_i]_\times \delta \boldsymbol{\theta} + \delta \mathbf{d}_f, \quad (19)$$

$$\begin{aligned} \mathbf{J} \delta \dot{\boldsymbol{\Omega}} = & - \sum_i [\mathbf{b}_i]_\times \mathbf{R}^\top \frac{T_i}{d_i} \mathbf{P}_i \delta \mathbf{p} + \sum_i \left(T_i [\mathbf{b}_i]_\times [\mathbf{R}^\top \mathbf{u}_i]_\times + [\mathbf{b}_i]_\times \mathbf{R}^\top \frac{T_i}{d_i} \mathbf{P}_i \mathbf{R}[\mathbf{b}_i]_\times \right) \delta \boldsymbol{\theta} \\ & + ([\mathbf{J}\boldsymbol{\Omega}]_\times - [\boldsymbol{\Omega}]_\times \mathbf{J}) \delta \boldsymbol{\Omega} + \delta \mathbf{d}_\tau, \end{aligned} \quad (20)$$

plus $\delta \dot{\mathbf{p}} = \delta \mathbf{v}$, $\delta \dot{\boldsymbol{\theta}} = -[\boldsymbol{\Omega}]_\times \delta \boldsymbol{\theta} + \delta \boldsymbol{\Omega}$.

Cable-length update (tension-based localisation). Each taut cable ($T_i > 1$ N) gives the scalar measurement $z_i = L_i(1 + T_i/EA) = \|\mathbf{a}_i - \mathbf{p} - \mathbf{R}\mathbf{b}_i\|$ with

$$\mathbf{H}_i = [-\mathbf{u}_i^\top \quad \mathbf{0} \quad \mathbf{u}_i^\top \mathbf{R}[\mathbf{b}_i]_\times \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}], \quad (21)$$

i.e. row i of $-\mathbf{W}^\top$ in body coordinates: the eight lengths observe the full pose whenever $\mathbf{W}$ has rank 6. The anchors come from the robots' own filters, so the measurement variance includes the anchor error.

Tag update. A camera on robot k measures the platform pose in its own frame, $(\mathbf{p}_{\text{rel}}, \mathbf{R}_{\text{rel}})$; with the robot's estimated pose $\mathbf{p}_{\text{tag}} = \hat{\mathbf{p}}_k + \hat{\mathbf{R}}_k \mathbf{p}_{\text{rel}}$, $\mathbf{R}_{\text{tag}} = \hat{\mathbf{R}}_k \mathbf{R}_{\text{rel}}$, and the residuals are $\mathbf{p}_{\text{tag}} - \mathbf{p}$ and $\log(\mathbf{R}^\top \mathbf{R}_{\text{tag}})^\vee$. An attitude error ϵ of a robot at range ℓ moves the tag position by $\ell\epsilon$ (7 mm for 0.1° at 4 m), so the tags are weighted well below the cable lengths; they guard against a wrong mode rather than set the precision.

Why not length control alone. A length-controlled cable turns an anchor position error δa along the cable into a tension error

$$\delta T = \frac{EA}{L} \delta a \quad (34 \text{ N/mm at } L = 2.9 \text{ m}), \quad (22)$$

so anchors known to a few millimetres cannot be combined with eight stiff length servos. With four tension-mode cables the four length-mode cables only fix four of the six platform degrees of freedom, an anchor error moves the platform instead of loading the cables, and the tensions stay near their set-points. This is the estimation side of the hybrid scheme of Section 5.2.

5 Control

5.1 Platform tracking on SE(3)

Reference $(\mathbf{p}_d, \mathbf{v}_d, \mathbf{a}_d, \mathbf{R}_d, \boldsymbol{\Omega}_d)$. Errors

$$\mathbf{e}_p = \mathbf{p}_d - \mathbf{p}, \quad \mathbf{e}_v = \mathbf{v}_d - \mathbf{v}, \quad \mathbf{e}_R = \frac{1}{2}(\mathbf{R}_d^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_d)^\vee, \quad \mathbf{e}_\Omega = \boldsymbol{\Omega} - \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\Omega}_d. \quad (23)$$

The wrench the cables must supply (`PlatformController::step`) is

$$\mathbf{f} = m \left(\mathbf{a}_d + k_p \mathbf{e}_p + k_d \mathbf{e}_v + k_i \int \mathbf{e}_p + g \mathbf{e}_3 \right) - \mathbf{f}_k - \hat{\mathbf{d}}_f, \quad (24)$$

$$\boldsymbol{\tau} = \mathbf{R} \left[\mathbf{J} \left(-k_p \mathbf{e}_R - k_d \mathbf{e}_\Omega - k_i \int \mathbf{e}_R - \boldsymbol{\Omega} \times \mathbf{R}^\top \mathbf{R}_d \boldsymbol{\Omega}_d \right) + \boldsymbol{\Omega} \times \mathbf{J} \boldsymbol{\Omega} \right] - \boldsymbol{\tau}_k - \mathbf{R} \hat{\mathbf{d}}_\tau, \quad (25)$$

with $k_p = \omega_n^2$, $k_d = 2\zeta\omega_n$, $k_i = 0.5\omega_n^3$ ($\omega_n = 7$ rad/s), bounded integrals, and the observer's disturbance estimate fed forward. $\mathbf{w} = (\mathbf{f}, \boldsymbol{\tau})$ goes through (15).

5.2 Hybrid winch commands

Drone winches run in tension mode (6) and ground winches in length mode (5); all get $T^{\text{ff}} = \mathbf{t}$ from (15).

Length-mode cables servo to the inverse kinematics of a corrected pose, $L_i^c = d_i(\mathbf{p}_d + k_{ik} \int \mathbf{e}_p, \mathbf{R}_d \exp[-k_{ik} \int \mathbf{e}_R]_{\times}, \hat{\mathbf{a}}_i)/(1 + t_i/EA)$: in the directions these cables hold stiffly a wrench integral has almost no authority, so the steady error is removed kinematically. They also carry a slow admittance on their load cells,

$$L_i^c \leftarrow L_i^c + \delta L_i, \quad \delta \dot{L}_i = k_a (T_i - t_i), \quad |\delta L_i| \leq \delta_{\max}, \quad (26)$$

so each is a stiff length servo above the bandwidth of (26) and follows its tension set-point below it. This is what makes the scheme work with *estimated* anchors: measured on the model, a millimetre of ground-anchor error shifts a ground-cable tension by about 4 N, and (26) reduces the tension ripple from 2.2 to 0.7 N. With the true state it is switched off.

Tension-mode cables get the rate of a commanded twist,

$$\dot{L}_i^c = \mathbf{u}_i^{\top} (\dot{\hat{\mathbf{a}}}_i - \mathbf{v}_c - \boldsymbol{\omega}_c \times \mathbf{r}_i), \quad \mathbf{v}_c = \mathbf{v}_d + k_t \mathbf{e}_p, \quad \boldsymbol{\omega}_c = \mathbf{R}_d \boldsymbol{\Omega}_d - k_t \mathbf{R} \mathbf{e}_R, \quad (27)$$

with the correction limited to 0.05 m/s.

Why (27). Take one translational direction held by tension-mode cables with projection γ_i , total damping $c = \sum_i \gamma_i^2 c_w$. With $\dot{L}^c$ equal to the reference rate only ($k_t = 0$), (6) and the PID give $c \dot{e} + m k_p e + m k_i \int e = 0$ (the platform inertia is negligible against c), i.e.

$$\omega = \sqrt{m k_i / c}, \quad \zeta = \frac{m k_p}{2 \sqrt{m k_i c}}. \quad (28)$$

With $m = 3$ kg, $k_i = 171.5$, $k_p = 49$, $c \approx 1800$ Ns/m: 0.085 Hz and $\zeta = 0.076$. The linearised closed-loop map of the full model (Section 8) has exactly this mode: 0.08 Hz, $\zeta = 0.089$. With (27) the same equation becomes $c(\dot{e} + k_t e) + m k_p e + m k_i \int e = 0$, with real poles near $-k_t$ and $-m k_i / (c k_t + m k_p)$: the error decays at $k_t = 3.5$ s⁻¹ and the oscillation is gone.

5.3 Quadrotor

Geometric tracking (Lee, Leok, McClamroch 2010) with the cable force fed forward from the load cell and the estimated cable direction, $\hat{\mathbf{F}}_c = -T \hat{\mathbf{u}} - \frac{1}{2} \rho g L \mathbf{e}_3$ (DroneController):

$$\mathbf{F} = -\mathbf{K}_p(\mathbf{p}_q - \mathbf{p}_r) - \mathbf{K}_d(\mathbf{v}_q - \mathbf{v}_r) - \mathbf{K}_i \int (\mathbf{p}_q - \mathbf{p}_r) + m_q(\mathbf{a}_r + g \mathbf{e}_3) - \hat{\mathbf{F}}_c, \quad (29)$$

$$\mathbf{f} = \mathbf{F} \cdot \mathbf{R}_q \mathbf{e}_3, \quad \mathbf{b}_3 = \mathbf{F} / \|\mathbf{F}\|, \quad \mathbf{R}_c = [\mathbf{b}_2 \times \mathbf{b}_3, \mathbf{b}_2, \mathbf{b}_3], \quad \mathbf{b}_2 = \frac{\mathbf{b}_3 \times \mathbf{b}_{1c}}{\|\mathbf{b}_3 \times \mathbf{b}_{1c}\|}, \quad (30)$$

$$\mathbf{M} = -\mathbf{K}_R \frac{1}{2} (\mathbf{R}_c^{\top} \mathbf{R}_q - \mathbf{R}_q^{\top} \mathbf{R}_c)^{\vee} - \mathbf{K}_{\Omega} \boldsymbol{\Omega}_q + \boldsymbol{\Omega}_q \times \mathbf{J}_q \boldsymbol{\Omega}_q - \mathbf{r}_a \times \mathbf{R}_q^{\top} \hat{\mathbf{F}}_c, \quad (31)$$

with the horizontal part of $\mathbf{F}$ limited to $F_z \tan \theta_{\max}$. The last term cancels the cable moment about the centre of mass in (9).

5.4 Ground robot

Desired planar velocity $\mathbf{v}^* = \dot{\mathbf{g}} + k(\mathbf{g} - \mathbf{x})$ towards the formation goal $\mathbf{g}$, limited to 0.6 m/s; heading error e_{θ} to $\mathbf{v}^*$ (reversing if that turns less); $v = \|\mathbf{v}^*\| \cos e_{\theta}$, $\omega = k_{\theta} e_{\theta}$; acceleration-limited; wheel speeds $(v \mp \frac{b}{2} \omega) / r$ (UgvController).

6 Perception and planning

Lidar and map (`lidar_scan`, `OccupancyGrid`). Each ground robot’s 2-D lidar returns ranges r_k at bearings α_k . A scan taken at the robot’s *estimated* pose updates a log-odds grid: cells along the beam $\ell \leftarrow \ell - 0.2$, the end cell $\ell \leftarrow \ell + 0.9$, $|\ell| \leq 4$; a cell is occupied for $\ell > 0.7$. A two-pass chamfer transform gives the distance field $D(\mathbf{x})$ to the nearest occupied cell.

Team validity (`TeamPlanner::valid`). With the formation centred on the platform at $\mathbf{c}$ and ground robots at $\mathbf{c} + \mathbf{o}_k$, the position is valid iff

$$D(\mathbf{c}) \geq r_b, \quad D(\mathbf{c} + \mathbf{o}_k) \geq r_r, \quad D(\mathbf{c} + s\mathbf{o}_k) \geq r_c \quad \forall s \in (0, 1), \quad k = 1..4, \quad (32)$$

each radius enlarged by 8% (chamfer error) plus one cell. The map is two-dimensional, so the team does not pass a cable over an obstacle, and unmapped space counts as free.

Search. A* over the grid (8-connected, octile heuristic) with lazily cached validity, then line-of-sight shortcutting; it is repeated as the map grows.

Three dimensions (`depth_scan`, `ElevationGrid`, `TeamPlanner3`). A downward depth camera on each drone gives points on whatever is below; the map keeps the highest return per cell, $H(\mathbf{x})$, and a lidar return raises its cell to at least the height of the scan plane. Dilating H by the robot, cable and tool radii gives H_r, H_c, H_b . A platform position $\mathbf{c} = (x, y, z)$ with the formation centred on it is valid iff

$$H_b(x, y) + \delta \leq z - \ell_{\text{tool}}, \quad H_r((x, y) + \mathbf{o}_k) \leq h_{\text{step}}, \quad (33)$$

$$H_c(\mathbf{q}_k(s)) + \delta \leq z_k(s) \quad \forall s \in [0, 1], \quad \text{all eight cables}, \quad (34)$$

where $\mathbf{q}_k(s), z_k(s)$ run along cable k from its platform corner to its fairlead (0.35 m on a ground robot, 4.5 m on a drone) and δ is the vertical clearance. The search is A* over (x, y, z) (26-connected) with the metric $\|\Delta xy\| + c_z|\Delta z|$, so the team lifts the payload over a low obstacle while the ground robots straddle it, and goes around a tall one. Cells within 0.3 m of the start are passable, so a start that new measurements have just made invalid can be left.

Reconfiguration (`TeamPlanner3::plan4`). The ground robots’ ring may be scaled by $\sigma \in \{0.6, \dots, 1\}$, so the state is (x, y, z, σ) and the robots sit at $(x, y) + \sigma\mathbf{o}_k$. A (z, σ) pair is admitted only if the tension distribution of Section 3.4 meets the weight exactly with every tension above $t_{\min} + 0.5$ N (tabulated once: level platform, formation centred). Moves are the 26 spatial neighbours at constant σ and a change of σ in place, costing $c_\sigma|\Delta\sigma|$. While the plan is followed the ring radius and the height change continuously, and the winches follow with the cable lengths from the inverse kinematics.

7 Keeping the cables taut

Three things were needed.

The hold region must be wrench-feasible. The formation holds position while the platform is inside a radius $R(z)$ and that radius was computed with a test that checked $t_{\min} \leq \mathbf{t} \leq t_{\max}$ but not $\mathbf{W}\mathbf{t} = \mathbf{w}$. With cables pinned at a bound the others cannot always balance the wrench, so the platform was allowed to go where a ground cable has to fall below $t_{\min}$. With the residual checked ($\|\mathbf{W}\mathbf{t} - \mathbf{w}\| < 0.05$ N), the feasible radius at $z = 1.3$ m is 0.40 m, not 0.88 m.

Reference governor (`PlatformController::governed`). The commanded reference is followed by $\dot{\mathbf{v}}_g = \text{sat}_{a_{\max}}(\mathbf{v}_d + k_g(\mathbf{p}_d - \mathbf{p}_g) - \mathbf{v}_g)/\Delta t$, $\dot{\mathbf{p}}_g = \mathbf{v}_g$ (and the same on $\text{SO}(3)$), so a velocity step in the command is not a tension impulse.

Slack guard. A length-mode cable measured below t_g is reeled in at $k_g^T(t_g - T_i)$ on top of (26).

8 Stability

8.1 Platform loop with ideal wrench

If the cables deliver $\mathbf{w}$ exactly and $\hat{\mathbf{d}} = \mathbf{d}$, the translational error obeys $\ddot{\mathbf{e}}_p + k_d \dot{\mathbf{e}}_p + k_p \mathbf{e}_p + k_i \int \mathbf{e}_p = 0$, Hurwitz iff $k_d k_p > k_i$ (here $12.6 \cdot 49 \gg 171.5$). For the attitude, with $\Psi = \frac{1}{2} \text{tr}(\mathbf{I} - \mathbf{R}_d^\top \mathbf{R})$ and $V = \frac{1}{2} \mathbf{e}_\Omega^\top \mathbf{J} \mathbf{e}_\Omega + k_p \lambda_J \Psi + c_1 \mathbf{e}_R^\top \mathbf{J} \mathbf{e}_\Omega$, the argument of Lee et al. gives $\dot{V} \leq -\mathbf{z}^\top \mathbf{Q} \mathbf{z}$, $\mathbf{z} = (\|\mathbf{e}_R\|, \|\mathbf{e}_\Omega\|)$, $\mathbf{Q} \succ 0$ for small c_1 , on the sublevel set $\Psi < 2$: exponential convergence from any initial attitude error below 180° .

8.2 What the ideal analysis leaves out

The wrench is produced through (15), the winch servos (5)–(6), elastic cables and robots that are themselves feedback systems, at three sample rates. The time scales are separated (cable axial mode 23 Hz, winch length servo 8 Hz, drone attitude about 3 Hz, platform loop 1.1 Hz, drone position 0.4 Hz) but not by enough for a clean singular-perturbation proof, and the hybrid winch scheme adds the slow mode analysed in Section 5.2.

8.3 Linearised closed-loop map

The sampled closed loop is therefore checked on the full model. Let Φ be the map of the closed-loop state (plant states, controller integrators) over one common control period $T_c = 20$ ms (20 physics steps; all three controllers fire at its start). `Simulator::monodromy` computes $\mathbf{A} = \partial \Phi / \partial \mathbf{x}$ at an equilibrium by central differences in error-state coordinates ($\delta \boldsymbol{\theta}$ for rotations). The equilibrium is locally exponentially stable iff the spectral radius $\rho(\mathbf{A}) < 1$; an eigenvalue μ corresponds to the continuous pole $s = \ln \mu / T_c$. The ground robots are parked for this test: a differential-drive base cannot be asymptotically stabilised to a point by smooth feedback (Brockett), and its go-to-goal law has a dead band, so its position is a neutral direction by design.

9 Results

9.1 Model against the Python reference (`tests/core_test.py`, part 1)

Structure matrix: maximum difference 3×10^{-16} over 300 random poses. Tension distribution: identical to the closed form to 2×10^{-12} N when no bound is active; projected gradient of the bounded solutions below 10^{-13} . Drone caps equal to the Python implementation. Forward kinematics recovers a pose to 10^{-9} m.

9.2 Closed loop on the C++ plant (part 2)

Lift 0.3 m with 10° roll and 15° yaw, then a 22 N payload.

	true state	sensors and estimators in the loop
position error held after the move (mean / max)	0.5 / 0.8 mm	0.8 / 1.1 mm
attitude error (max)	0.10°	0.26°
under the payload (mean / peak)	0.3 / 2.7 mm (announced)	1.7 / 8.6 mm (not announced)
cable tensions	6.7–35.5 N	5.2–36.9 N
platform estimate error (mean / max)	–	1.3 / 4.1 mm, 0.78° max
payload estimated by the observer	–	–22.0 N (true –22)

The full loop (plant, sensors, estimators, controllers at 1 kHz) takes 10–13 μ s per step, about 80 times real time on one core.

9.3 Stability (part 3)

Linearised map over 20 ms at the level pose, all loops closed, ground robots parked, 122 states.

	$\rho(\mathbf{A})$	slowest pole	least-damped mode
original hybrid scheme ($k_t = k_{ik} = k_a = 0$)	0.99990	-0.005 s^{-1}	0.08 Hz, $\zeta = 0.089$
with commanded twist (27) and pose-integral inverse kinematics	0.99823	-0.089 s^{-1}	22.8 Hz, $\zeta = 0.15$
as deployed (plus admittance (26))	0.99944	-0.028 s^{-1}	22.9 Hz, $\zeta = 0.15$

All eigenvalues are inside the unit circle. The -0.089 s^{-1} pole is the drones’ position integrator (k_i/k_p); the -0.028 s^{-1} pole is the admittance redistributing the ground-cable lengths; the 23 Hz mode is the cables’ axial mode. This is a local result at one equilibrium, for the model, with the true state fed back; it is not a proof for the estimator-in-the-loop system or for large motions.

9.4 In Isaac Sim

`tests/isaac_physics_test.py`: the same manoeuvre, then a 1 m translation of the whole formation; statistics over every physics step after 1 s. The position error is measured against the commanded reference, so for the C++ core it includes the lag of the reference governor at the two command steps (the maxima).

	Python controllers, true state	C++ core, true state	C++ core, sensors in the loop
position error rms / max	4.0 / 11.4 mm	2.1 / 13.2 mm	3.0 / 14.2 mm
attitude error mean	0.71°	0.26°	0.77°
a cable slack	0.55% of the time	never	never
lowest tension	0 N	1.4 N	3.0 N
platform estimate error mean / max	–	–	1.3 / 2.8 mm

`run_traj.py --scenario track --core` (figure-eight, 48 s, sensors in the loop): 4.4 mm mean, 15.9 mm max; a cable slack 0.15% of the time (about 70 ms in total).

9.5 Sensor-based planning

`tests/core_plan_test.py` (2-D, lidar only, no prior map, replanning every 0.5 m): goal reached in 34 plans of about 3 ms; path 16.5 m, around both obstacles; true clearances 0.55 m for the ground robots (0.55 required) and for the cables (0.20).

`tests/core_plan3d_test.py` (elevation map from depth cameras and lidars): goal reached in 31 plans of about 21 ms; path 15.4 m, around the cabin and over the pallet at $z = 1.5$ m with the ground robots straddling it; true clearances 0.81 m for the ground robots, 0.32 m for the cables, 0.22 m for the tool.

`run_traj.py --scenario plan3d --core` (the same planner running in Isaac while the team moves, sensors and estimators in the loop): goal reached, 15.4 m in 57 s, 31 plans of 21 ms; tracking error 3.1 mm mean, 25 mm max at the corners of the plan; recorded clearances 0.80 m for the ground robots and 0.34 m for the cables; a cable slack 0.04% of the time. The first attempt drove a ground robot into the stack of blocks on the site, which was not among the obstacles given to the range models; it is now.

`run_traj.py --scenario course --core` (reconfiguration; a 4.4 m gate in a 2 m wall for a team 5.0 m wide, a crate, two 3 m pillars; search over $(x, y, z, \text{ring scale})$ with the wrench-feasible (height, scale) pairs tabulated beforehand): goal reached, 19.4 m in 68 s, 39 plans of 92 ms. The ground robots' ring goes from 2.75 m to 1.65 m at the gate and back, and the ground cables from 2.96 m to 1.74 m (drone cables 3.49–3.62 m). Recorded clearances: ground robots 0.80 m, cables 0.80 m, drones 1.73 m. Tracking error 2.8 mm mean, 25 mm max; no cable slack (lowest tension 0.4 N).

9.6 Limits

The sensors are measurement models on the true simulator state: ranges are cast against the collider boxes, tag poses and fixes are the true values with noise; no images or point clouds are processed and there is no latency. The sensors do not see the team itself. The platform has no inertial sensor, so its attitude estimate ($0.3\text{--}0.9^\circ$) is the weakest part. The planner assumes the formation centred on the platform and a level platform, and treats unseen space as free. The cables are never slack in the manoeuvre test but still are for a few tens of milliseconds in the two long trajectory runs.